

Rapid, high-efficient and scalable exfoliation of high-quality boron nitride nanosheets and their application in lithium-sulfur batteries

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ABSTRACT

Boron nitride nanosheets (BNNs) have gained significant attraction in energy and environment fields because of their two-dimensional (2D) nature, large band gap and high thermal/mechanical performance. However, the current low production efficiency of high-quality BNNs is still a bottleneck limiting their applications. Herein, based on sonication-assisted liquid-phase exfoliation, we demonstrated a rapid, high-efficient and scalable production strategy of BNNs and documented the effects of a spectrum of exfoliation factors (e.g., ultrasonic condition, solvent and bulk material feeding) on the yield of BNNs. A record of yield of 72.5% was achieved while the exfoliated BNNs have few-layer and defect-free feature. Thanks to the Lewis acid sites of the boron atoms, the BNNs can interact with the polysulfide anions in liquid electrolyte and also can facilitate the uniform lithium deposition, which finally endow a lithium-sulfur (Li-S) battery with long life. This work provides a facile and rapid strategy for large scale preparation of high-quality BNNs, also contributes a long-life strategy for dendrite-free Li-S battery, opens new avenues of BNNs in energy application.

KEYWORDS

boron nitride nanosheets, cavitation, sonication-assisted liquid-phase exfoliation (SALPE), modified separators, lithium-sulfur batteries

1 Introduction

Boron nitride nanosheets (BNNs), two-dimensional (2D) wide-bandgap (~ 6 eV) nanomaterials, have received numerous attentions because of their ultrahigh thermal conductivity of $300\text{--}2,000\text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ [1, 2], highly electrical insulation [3], excellent chemical/thermal stability [4] and mechanical strength [5]. Owing to the atomic-layer (single or few-layers) thickness and 2D morphology, BNNs have been considered as ideal candidates for thermal management [6–8], high voltage insulation [9], energy storage [10, 11], and photoelectrochemical applications [12]. Despite high demand, the production of high-quality BNNs is still limited by the low efficiency [13].

So far, significant efforts have been made to fabricate BNNs. Typical methods include chemical vapor deposition (CVD) [14, 15], mechanical ball milling [16, 17], electrochemical exfoliation [18], hydrothermal synthesis [19], microfluidic shear peeling [20], supercritical fluid stripping [21], ion intercalating methods [22] etc. However, most of these approaches need a long production period (typically > 5 h), and/or cause imperfections of the BNNs. In contrast, liquid phase exfoliation (LPE) is a facile technique to produce defect-free 2D nanosheets [23], and it has been used to prepare BNNs from bulk hexagonal boron nitride (h-BN) flakes. Nevertheless, the LPE output of BNNs from bulk h-BN is still low (typically < 0.3 g). In addition, the production also suffers

from low yield (typically $< 20\%$), volatile of toxic organic solvents and high energy consumption [24–27].

Lithium-sulfur (Li-S) batteries have high theoretical specific capacity ($1,675\text{ mAh}\cdot\text{g}^{-1}$) and energy density ($2,600\text{ Wh}\cdot\text{kg}^{-1}$), which are one of most promising devices for high capacity energy-storage demands [28–30]. However, many issues result in low Coulombic efficiency (CE) and security flaws of Li-S batteries, typically including shuttle effect of soluble long-chain lithium polysulfides (LiPSs), and Li dendrites growth in cycling [31, 32]. Hence, functional materials which can greatly control the LiPSs shuttling while also effectively protect the Li metal anodes are highly desirable [33]. Typically, the h-BN materials possess high Young's modulus (36.5 GPa) [34] and lithiophilic surface, denoting it a robust block material to restrain Li dendrites without compromising Li^+ conductivity. In addition, the p_z orbital of B atoms can act as Lewis acid sites to catch the polysulfides anions in electrolyte, enhancing the capacity of Li-S batteries (for detailed discussion in Fig. S1 in the Electronic Supplementary Material (ESM)) [35]. To date, the h-BN hybrid materials, such as the functionalized BNNs/graphene interlayer [36] and the functionalized BNNs modified separator [37] have been applied in Li-S batteries. However, these positive and negative charged h-BN interlayer or modified layer inhibit the shuttle effect greatly relying on strong ion repelling effect of these functional groups as well as synergetic effect of graphene instead of the intrinsic properties

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of the h-BN. Moreover, the addition of large amount of graphene (> 60 wt.%) increases the probability of internal short-circuit risk and a long preparation time for functionalization is needed, which is not favorable for scalable production [36, 38]. Since pure few-layer BNNs possess extremely high specific surface area and Young's modulus (950 GPa) [39], they can provide much more Lewis acid sites in their 2D form at a low loading level as well restrain Li dendrites formation during cycling, and easily for scalable production compare to functionalization. Therefore, introducing the high-quality pure BNNs is highly desirable for significantly improving the cyclability and rate capability of Li-S batteries.

Herein, we report a rapid and efficient sonication-assisted liquid-phase exfoliation (SALPE) method for scalable preparation of high-quality BNNs. A record of total yield of 72.5% was achieved while the exfoliated BNNs also have few-layer and defect-free feature. In addition, the as-prepared BNNs show a narrow diameter and thickness distribution and without any doping and functionalization even after 8 recycles, showing stable solutions of high concentration of 15 mg·mL⁻¹. Furthermore, our work also demonstrated that the BNNs own a great prospect in electrochemical energy storage application. Using the BNNs modified polypropylene (PP) separator, we show that the Li-S batteries exhibit superior cyclability and rate capability.

2 Experimental

2.1 Materials

The bulk h-BN flake powder (CFP003) was obtained from 3M Technical Ceramics (U.S.A.). The isopropyl alcohol (IPA) (99%) and N-methylpyrrolidinone (NMP) were supplied by Adamas Reagent (China). Polyvinylidene fluoride (PVDF) was provided by SOLVAY (Belgium). Lithium metal foils (15.6 mm × 0.45 mm), Ketjen black (KB) and Celgard 2500 were purchased from Guangdong Canrd New Energy Technology Co., Ltd (China). The electrolyte composed of 1 M bis(trifluoromethane) sulfonamide lithium (LiTFSI) in (1,3-dioxolane (DOL) and 1,2-dimethoxyethane (DME) = 1:1 v/v) with 1 wt.% LiNO₃ was supplied by DodoChem (Suzhou, China).

2.2 Exfoliation of BNNs

A certain amount (0.5–3.0 g) of bulk h-BN flake powder was added into an IPA/water mixture solvent (100 mL) with different composition. The solutions were sonicated in the ultrasonic bath (Shanghai ShengXi Ultrasonic Instrument, FS-450N, 20 kHz, 30 kHz, 40 kHz) for 80–240 min with different ultrasonic power in ice water bath. After centrifugation for 7 min at 3,000 rpm, the supernatants were collected to obtain the exfoliated BNNs. The precipitates were collected for the next exfoliation cycle.

2.3 Preparation of separators coated with bulk h-BN and BNNs

200 mg mixture of bulk h-BN flake powder (or BNNs) and PVDF binder (*m/m* = 8:2) into 1 mL NMP solvent. After magnetic stirring for 3 h, the slurries were casted onto one side of commercial PP separators by a doctor blade and dried at 50 °C for 12 h in a vacuum oven. Finally, the PP-BN and PP-BNNS functional separators were obtained and punched into a size of Φ 19 mm before used.

2.4 Preparation of KB/S cathodes

The KB/S composite was prepared via melt-diffusion method.

The KB and S with a ratio of 3:7 (*m/m*) mixed and heated in a sealed glass bottle at 155 °C for 6 h. The KB/S cathodes were prepared by blade-casting method. The slurry was prepared by mixing KB/S and PVDF with a mass ratio of 9:1 in NMP and stirred for 3 h. Then, cast the slurry onto the carbon-coated Al foil and dried in vacuum oven at 50 °C for 12 h. Finally, the cathodes were obtained and cut into circular disk with a diameter of 12 mm. The average areal sulfur mass loading is 2 mg·cm⁻².

2.5 Characterization

The scanning electron microscope (SEM) images of the bulk h-BN, BNNs, the PP, PP-BN and PP-BNNS separators, KB/S cathode and Li foils and the cross-sectional SEM images of PP-BN and PP-BNNS separators were observed by a SEM (Nova NanoSEM 450, FEI). The transmission electron microscope (TEM) and atomic force microscope (AFM) images of the BNNs were also obtained by a transmission electron microscope (TALOS F200X, FEI) and an atomic force microscope (Multimode Nanoscope IIIa), respectively. The structural information of the bulk h-BN, BNNs and KB/S was collected via Fourier transform infrared spectroscopy (FTIR, Paragon 1000, PerkinElmer), X-ray photoelectron spectroscopy (XPS, AXIS UltraDLD, Kratos) and X-ray diffraction (XRD, D8 Advance, Bruker). The PP, PP-BN and PP-BNNS separators were heated on a hot plate (DB-4A, Jintan Chenghui Instrument). The surface temperatures of these separators were tested and recorded by an infrared thermograph (FOTRIC-226). The mechanical properties of separators were analyzed by using the universal material tester (3343, INSTRON). The contact angle tester (DSA100, Kruss) was used for the contact angle images of the PP, PP-BN and PP-BNNS separators. Thermal gravimetric analysis (NETZSCH TG209 F3) was conducted to measure the sulfur contents with a heating rate of 10 °C·min⁻¹ in N₂ atmosphere.

2.6 Electrochemical measurements

The electrochemical performance measurements were tested in CR-2032-coin cells. The Li-S cells were assembled by using KB/S as the cathode, lithium foils as the counter electrode and 1 M LiTFSI (DOL/DME = 1:1, v/v) with 1 wt.% LiNO₃ additive as the electrolyte in a glove box with Ar atmosphere (H₂O, O₂ < 0.1 ppm). In order to maintain the consistency, the ratio of electrolyte to sulfur (*E/S*) for KB/S was controlled as 20 μ L·mg⁻¹ in each cell. The rate and cyclic performances of the cells with these separators were tested on LANHE CT2001A (5 V 10/50 mA) at room temperature within a voltage window of 1.7–2.8 V (vs. Li/Li⁺). All of the specific capacities were calculated based on the mass of sulfur. The electrochemical impedance spectroscopy (EIS) and cyclic voltammetry (CV) tests were performed on the VersaSTAT 3F (Princeton Applied Research) workstation at frequency range from 0.01 Hz to 100 kHz and a scan rate of 0.2 mV·s⁻¹, respectively.

2.7 Ionic conductivity measurement

The ionic conductivities of the PP, PP-BN and PP-BNNS separators were determined by using EIS in the frequency range of 0.01–10⁵ Hz at an amplitude of 10 mV. Each separator was sandwiched by two stainless-steel (SS) electrodes in symmetric cell (SS/PP/SS, SS/PP-BN/SS and SS/PP-BNNS/SS). The ionic conductivities of the PP, PP-BN and PP-BNNS separators are calculated by the following equation

$$\sigma = \frac{l}{R_s \times S} \quad (1)$$

where σ , l , R_s and S are represented the ionic conductivity, the thickness of the separator, the bulk resistance and the area of SS electrode ($\Phi = 15.6$ mm), respectively.

3 Results and discussion

3.1 BNNSs exfoliation

Figure 1(a) illustrates the exfoliation process of BNNSs from the bulk h-BN flakes. In a liquid phase, sonication can trigger a cavitation phenomenon, which refers to the nucleating, growing, gathering ultrasonic energy and negative pressure caused collapsing of micrometer sized bubbles [40]. The strong shockwave and high local turbulence induced by the bubble collapse can overcome van der Waals and lip-lip interactions between the adjacent BN layers, resulting in exfoliation of bulk h-BN [41, 42]. As such, many factors can influence the yield of exfoliated BNNSs, including the ultrasonic conditions (frequency, power, time, cycles), solvents and the bulk h-BN concentration.

We first investigated the role of ultrasonic frequency in BNNSs exfoliation. Three frequencies were considered and Fig. 1(b) shows that at 20 and 30 kHz, the yield increases with the ultrasonic power and it reaches the maximum of about 20% at 30 kHz and 375 W. However, at 40 kHz, the yield is extremely low (about 1%) even at a high power (i.e., 315 or 375 W). On the other hand, the role of ultrasonic power in BNNSs exfoliation is closely associated with the ultrasonic frequency. At lower ultrasonic power (150 and 225 W), the yield is higher at 20 kHz, while at 30 kHz the yield becomes higher at higher ultrasonic power (315 and 375 W). This experimental result is consistent with the previous simulation work about ultrasonic cavitation effect, where it was concluded that under frequency < 40 kHz, the ultrasonic devices show increased cavitation effect

with the decrease of frequency [43]. At a lower frequency, the peeling force caused by the bubble collapse increases with the ultrasonic power, and thus the yield becomes higher with the increase of ultrasonic power. However, at a higher frequency (30 kHz), the cavitation effect is not strong until the power is high enough. This can explain why a higher power is preferred to exfoliate bulk h-BN at a higher frequency (i.e., 30 kHz). Although the increase of power can increase the BNNSs yield, the further increase of ultrasonic power causes serious heating and energy consumption. Therefore, 315 W was selected in the following experiment.

Figure 1(c) shows the BNNSs yield versus sonication time at 20 and 30 kHz. One can see that the yield first increases with the sonication time and then the yield reaches a plateau with the further increase of sonication time. The time to a yield plateau is 200 min for 20 kHz, while it is 160 min for 30 kHz. Despite no increase of yield, further increase of sonication time may introduce defects into the BNNSs and/or decreasing their size [44, 45]. Therefore, the optimum condition is set as 30 kHz, 160 min and 315 W, which is intended to obtain higher exfoliation efficiency and high quality of BNNSs.

Figure 1(d) demonstrates the BNNSs yield versus IPA volume fraction in a mixed solvent (IPA and water) and R_a values, where R_a refers to the solubility of nanomaterials in a solvent and can be calculated according to Hansen solubility parameter (HSP) theory [46], and the calculated R_a values are presented in the ESM. A smaller R_a represents a better dispersion capability of the solvent, which is consistent with the experimental result. Figure 1(d) suggests that the yield exhibit comparable high values when the IPA/water ratios are 40/60, 45/55 and 50/50. Since IPA is much expensive than water, 40/60 is thus selected in our work.

Figure 1(e) presents the influence of initial concentration (C_{BN}) of bulk h-BN flakes on the yield of BNNSs. It can be seen

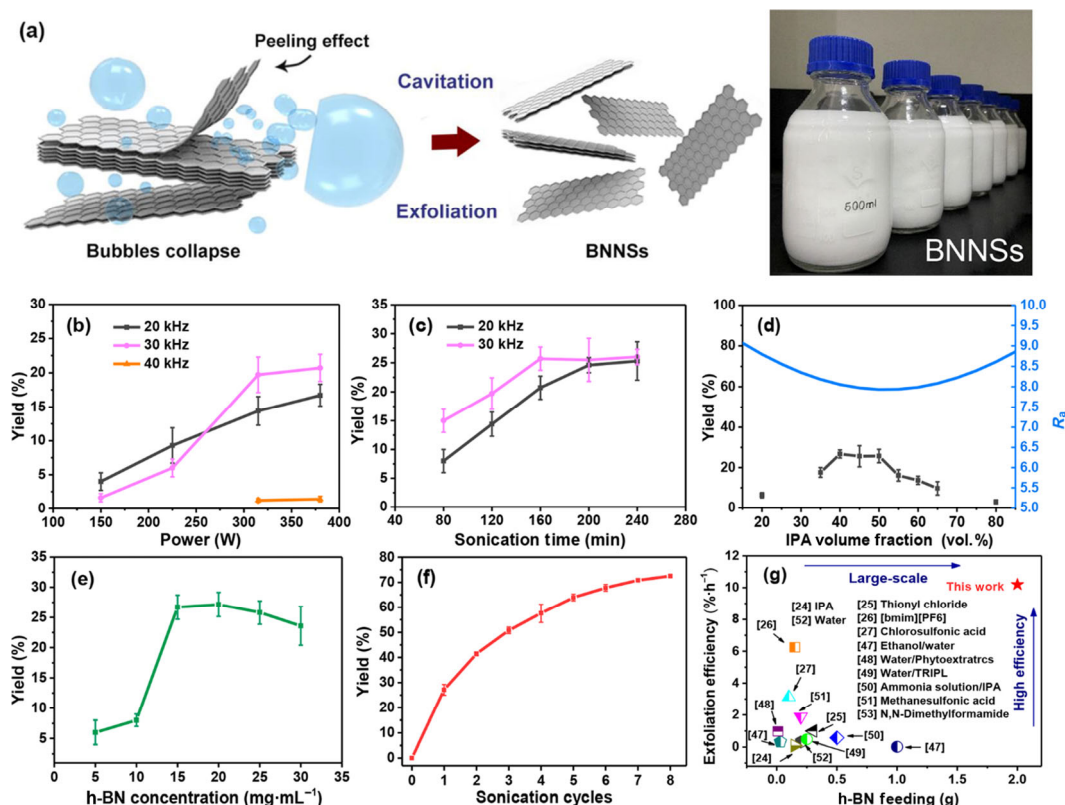


Figure 1 (a) Schematic illustration of the SALEP method for scalable preparation of BNNSs (left), and a digital photo of the BNNSs suspensions (right). The yield of BNNSs versus (b) ultrasonic frequency and power, (c) sonication time, (d) IPA volume fraction in the solvent, (e) h-BN concentration and (f) sonication cycles. (g) The comparison of h-BN feeding and exfoliation efficiency between our work and the reported SALEP methods.

that the yield increases rapidly as C_{BN} increases to $20 \text{ mg}\cdot\text{mL}^{-1}$, where it reaches the maximum of 27.15%. After that, the yield decreases slowly with further increase of C_{BN} . This is because the limited dispersion capability of the solvent. When the introduced bulk h-BN flakes exceed the critical concentration, the additional h-BN flakes result in precipitate, which decreases the exfoliation effect. Therefore, we set $20 \text{ mg}\cdot\text{mL}^{-1}$ as the optimal initial concentration of bulk h-BN flakes. The large-volume BNNs suspensions are shown in Fig. 1(a), revealing this SALPE method is scalable to meet the demand of high-quality BNNs.

When the BNNs were separated by centrifugation, the residual bulk h-BN flakes can be re-exfoliated. Figure 1(f) shows that the BNNs yield increases with the sonication cycle number, and the total yield eventually can reach 72.5% after 8 cycles. The SEM images of the prepared BNNs after different sonication cycles were provided in Fig. S2 in the ESM. One can see that the BNNs remain comparable morphology even after several sonication cycles and the average size of the BNNs only shows slight decrease, from $0.55 \mu\text{m}$ after the first sonication exfoliation to $0.47 \mu\text{m}$ after the eighth. The exfoliated BNNs can be well dispersed in IPA/water mixture at high concentrations of $15 \text{ mg}\cdot\text{mL}^{-1}$.

In order to demonstrate the merits of our method, exfoliation efficiency is introduced, which is defined as the ratio of BNNs yield to the exfoliation time. Figure 1(g) summarizes the h-BN feeding and exfoliation efficiency of BNNs in this work and other SALPE methods (see detailed information in Table S2 in the ESM) [24–27, 47–53]. One can see that our method exhibits both high exfoliation efficiency and production capacity, showing superiority in scalable production of BNNs. In addition, our method does not need large amount of expensive agent or introducing other functional groups (Fig. S3 and Table S3 in the ESM).

3.2 Characterization of exfoliated BNNs

Figures 2(a) and 2(b) provide typical SEM images of the bulk h-BN flakes and the exfoliated BNNs. The bulk h-BN flakes have a lateral size ranging from several hundred nanometers to a few micrometers. In contrast, the size of BNNs decreases and their lateral size is basically within $1.5 \mu\text{m}$ and the average value is $0.55 \mu\text{m}$ (Figs. S2(a) and S2(i) in the ESM). In addition, the BNNs are isolated each other, which are different from the bulk h-BN flakes. Folding and curling can also be observed at the BNNs edges, which are typical feature of few-layered BNNs.

TEM and AFM were further used to investigate the BNNs morphology. Figure 2(c) shows that the BNNs display a typical sheet-like shape and smooth edges. The selected-area electron diffraction (SAED) pattern of BNNs shows typical six-fold symmetry of h-BN (inset of Fig. 2(c), upper right), indicating that the exfoliation treatment does not affect the structure of h-BN. And the high-resolution TEM (HRTEM) image inset of Fig. 2(c) (lower right) clearly shows the edge of the BNNs consisting of 8 layers. Typical AFM images of the BNNs are presented in Figs. 2(d) and 2(e), suggesting the lateral size and the thickness of BNNs are about $0.55 \mu\text{m}$ and 2.9 nm . Considering the interlayer distance of BN is 0.333 nm [3], the BNNs are composed of about 8–9 layers, which is greatly consistent with the result of the HRTEM image inset in lower right of Fig. 2(c). These results greatly demonstrate the high-quality of the few-layered BNNs (~ 8 layers).

Figure 2(f) presents the FTIR spectra of bulk h-BN flakes and BNNs. One can see that there are two absorption peaks at $1,373$ and 817 cm^{-1} representing stretching vibration of B–N bond and bending vibration of B–N–B, respectively, confirming that the BNNs retain the lattice structure of h-BN [54]. Compared with the bulk h-BN flakes, the absorption peak of

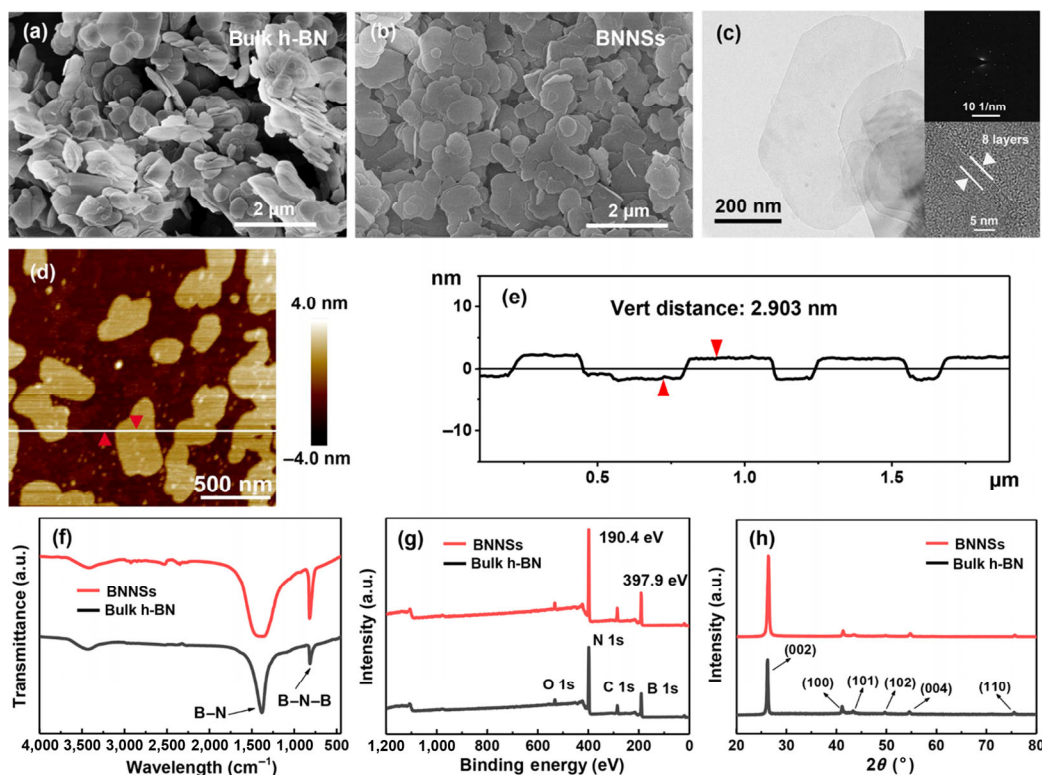


Figure 2 Characterization of the exfoliated BNNs. (a) and (b) SEM images of bulk h-BN flakes and the exfoliated BNNs. (c) TEM images of exfoliated BNNs and the inset shows the HRTEM image and the SAED pattern. (d) AFM image and (e) the corresponding height profile of the exfoliated BNNs. (f) FTIR spectra of bulk h-BN flakes and the exfoliated BNNs. (g) XPS survey of bulk h-BN flakes and the exfoliated BNNs. (h) XRD patterns of bulk h-BN flakes and the exfoliated BNNs.

BNNs at $1,373\text{ cm}^{-1}$ is broader and the relative peak intensity between B–N stretching vibration and B–N–B bending vibration decreases significantly, which should be ascribed to the thinner thickness of BNNs [55]. Except for the broad band at $3,422\text{ cm}^{-1}$ caused by O–H and/or water absorption, the spectrum of BNNs does not display any additional peaks, indicating that functional groups were not introduced during the exfoliation process.

Figure 2(g) gives the XPS patterns of the bulk h-BN flakes and BNNs, and Figs. S4(a) and S4(b) in the ESM show the high-resolution XPS scan of B (1s peak at around 397.9 eV) and N (1s at around 190.4 eV) [56]. As shown in Fig. S4(c) in the ESM, the B/N atomic ratio of BNNs is nearly 1:1, indicating their high purity. Figure 2(h) provides the XRD patterns of the bulk h-BN flakes and BNNs, which show typical peaks indexed with the crystalline planes (002), (100), (102), (004) and (110) of h-BN [57]. Compared with the bulk h-BN flakes, the relative intensity of (002) peak to other diffraction peaks was remarkably stronger in BNNs. This was probably because the exposed (002) crystal faces substantially increase in BNNs after exfoliation along the (002) plane.

3.3 BNNs-coated battery separators

Figure 3 presents SEM and optical images of BN coated PP separators. The BNNs coated separator (PP-BNNS) shows a smoother surface (Figs. 3(b) and 3(d)), while the bulk h-BN coated separator (PP-BN) exhibits agglomeration (Figs. 3(a) and 3(c)). Under the same blade coating conditions, the thickness of the coated BNNs ($1.06\text{ mg}\cdot\text{cm}^{-2}$) is about $34\text{ }\mu\text{m}$ (Fig. 3(d)), which is thicker than the coated bulk h-BN ($1.37\text{ mg}\cdot\text{cm}^{-2}$) (Fig. 3(c)). The uniformly distributed BNNs coating should be attributed to the decreased thickness of the exfoliated layers, which is expected to effectively entrap the polysulfides.

Thermal properties of PP-BN and PP-BNNS separators were compared with the commercial PP separator (Fig. 3(e)). After heated on a hot plate at $80\text{ }^{\circ}\text{C}$ for 1 min, the PP separator underwent complete crimp with the surface temperature of $44.1\text{ }^{\circ}\text{C}$, while the PP-BNNS separator remained intact with the surface temperature of $58.6\text{ }^{\circ}\text{C}$. The excellent thermal and chemical stability and ultrahigh thermal conductivity of h-BN can lead to the enhanced thermal stability and improved heat-transfer rate, which are expected to avoid the risk of internal short-circuit failures and thermal runaway in batteries. As shown in Fig. S5 and Table S4 in the ESM, the PP-BNNS separator exhibits the highest Young's modulus of 0.635 GPa ,

whereas the Young's modulus of the PP-BN and PP are 0.424 and 0.219 GPa , respectively, indicating the superiority of the PP-BNNS separator to restrain Li dendrites growth. To learn the effects of the modified layers on wettability, the dynamic contact angle measurements for the liquid electrolyte were performed for these three separators as depicted in Fig. S6 in the ESM. Interestingly, the PP-BNNS separator shows a smallest initial contact angle of 23.3° and quickly absorbed the droplet within 6 s compared with the pure PP and PP-BN separators. The much-improved wettability of the PP-BNNS should attributed to the polar B and N atoms. Combined high strength and high affinity to liquid electrolyte, the PP-BNNS separator can greatly enhance the cycling performance as well as protect the Li anodes.

Furthermore, the ionic conductivities of the PP, PP-BN and PP-BNNS separators were measured by using EIS measurements in symmetric cells (SS/PP/SS, SS/PP-BN/SS and SS/PP-BNNS/SS) at room temperature. The EIS curves of these symmetric cells are plotted in Fig. S7 in the ESM. The ionic conductivity of the PP-BNNS shows $1.35\text{ mS}\cdot\text{cm}^{-1}$, whereas the PP-BN and PP show conductivity of 1.24 and $0.95\text{ mS}\cdot\text{cm}^{-1}$, respectively, indicating that the BNNs coating layer can ensure the transportation of ionic in the electrolyte (i.e., 1 M LiTFSI (DOL/DME = 1:1 v/v)). These results clearly suggest that BNNs can improve the Li^{+} conductivity, possibly because B atoms in BNNs can act as Lewis acid sites. This may result in higher density of Li^{+} on BNNs surface by interacting with the TFSI anions in the electrolyte [58].

The Li/PP/Li, Li/PP-BN/Li and Li/PP-BNNS/Li symmetric cells were assembled and measured to investigate the stability against lithium metal anode during Li plating and stripping process. As shown in Fig. S8 in the ESM, the Li/PP-BNNS/Li cell shows the lowest voltage hysteresis ($\sim 150\text{ mV}$) in comparison with Li/PP-BN/Li and Li/PP/Li cells at a current density of $20\text{ mA}\cdot\text{cm}^{-2}$ over 500 h without obvious voltage changes, demonstrating that the PP-BNNS separator can protect lithium metal effectively. The enhanced cycling performance of the PP-BNNS separator in symmetric cell should be attributed to its favorable ionic conductivity, unique 2D structure and high Young's modulus [59]. As presented in Fig. S5 and Table S4 in the ESM, the PP-BNNS demonstrates highest Young's modulus (0.635 GPa). These results clearly show that the BNNs are robust enough to reduce the dendrites and can guide uniform Li deposition, which can greatly enhance the safety issues and cycling life of the cells.

CR-2032-coin cells were assembled using the KB/S and Li

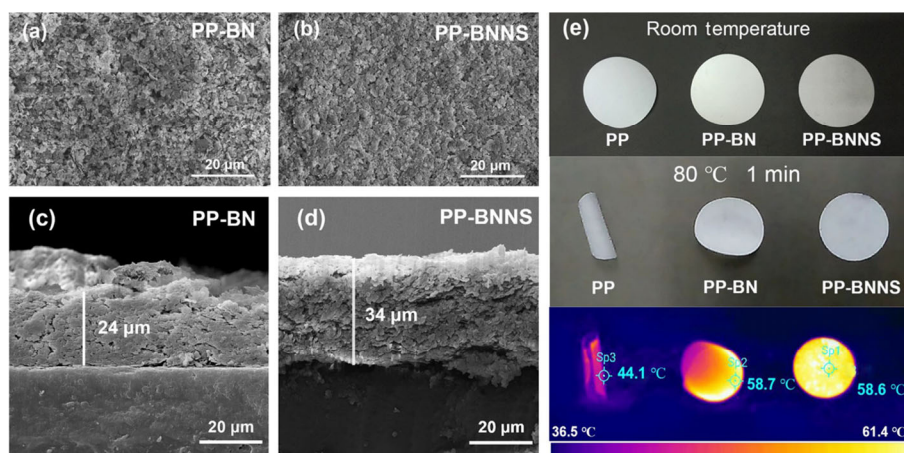


Figure 3 SEM images of (a) and (c) PP-BN separator, and (b) and (d) PP-BNNS separator. (a) and (b) for the top surface, and (c) and (d) for the cross-sections. (e) Thermal properties of PP, PP-BN and PP-BNNS separators. Samples were heated on a hot plate at $80\text{ }^{\circ}\text{C}$ for 1 min. Photographs were taken before and after heating. Infrared thermal image shows the temperature of samples after heating at $80\text{ }^{\circ}\text{C}$ for 1 min.

foils as the cathodes and anodes, respectively, to further evaluate the electrochemical performance of the PP, PP-BN and PP-BNNS separators in Li-S batteries. Figures S9(a) and S9(b) in the ESM confirm the existence of sulfur in the composite cathode and the actual sulfur content is about 70 wt.%. In addition, Figs. S9(c) and S9(d) in the ESM also suggest the uniform distribution of sulfur in KB and no obvious sulfur particle aggregation. Figure 4(a) discloses the galvanostatic charge/discharge (GCD) profiles of cells with the PP, PP-BN and PP-BNNS separators at the same current density (0.5 C). Each curve shows one charge plateau and two discharge plateaus, corresponding to the transformation from $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ to S_8 and conversion from S_8 to long-chain polysulfides (Li_2S_n , $4 \leq n \leq 8$) then to $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$, respectively. Figure 4(b) shows that the initial capacity of PP-BNNS is $1,405 \text{ mAh}\cdot\text{g}^{-1}$ at a current density of 0.2 C, which is higher in comparison with the PP-BN ($1,359 \text{ mAh}\cdot\text{g}^{-1}$) and the pure PP ($1,014 \text{ mAh}\cdot\text{g}^{-1}$). These results suggest a higher utilization of sulfur in PP-BNNS cell, and it is because the BNNSs barrier layer limits the shuttle effect of the LiPSs.

The PP-BNNS cell was measured at different current rates to further investigate the advantage of BNNSs. Figure 4(b) shows the discharge capacities of the PP-BNNS are 946.4 and $818.4 \text{ mAh}\cdot\text{g}^{-1}$ at 0.5 C and 1 C , respectively, which are higher in comparison with the PP-BN and the pure PP under the same loading of the cathode. In addition, the CV curves of these cells show similar shape within two cathodic peaks at 2.31 and 2.02 V , corresponding to the conversion from S_8 to long-chain LiPSs and finally to $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$, respectively, and within two anodic peaks at 2.32 and 2.41 V , representing transformation of $\text{Li}_2\text{S}_2/\text{Li}_2\text{S}$ to long chain LiPSs and S_8 , respectively. These results are in consistent with their GCD profiles (Fig. 4(a)). Interestingly, the CV curve area of PP-BNNS is narrower and sharper in comparison with PP-BN and PP at the scan rate of $0.2 \text{ mV}\cdot\text{s}^{-1}$, indicating good reversible redox conversion and high utilization of sulfur (Fig. 4(c)).

The EIS measurements were carried out to examine the

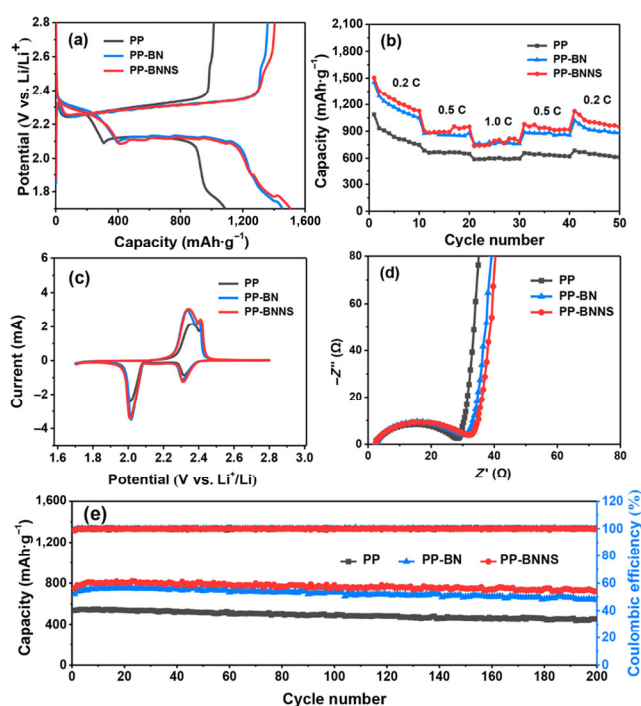


Figure 4 Electrochemical performances of cells with PP, PP-BN and PP-BNNS separators. (a) GCD profiles at the current density of 0.5 C. (b) Rate performances. (c) CV curves. (d) Nyquist plots. (e) Cycling capacities and CE at the current density of 0.5 C.

interfacial resistances of Li-S batteries with different separators (Fig. 4(d)). Compared with the PP separator, PP-BN and PP-BNNS separators exhibit relatively larger electrical resistance about 5Ω due to the high electrical insulation of bulk h-BN or BNNSs. On the other hand, this result clearly shows that Li^+ can pass through the PP-BN and PP-BNNS easily and a low interfacial electrical resistance is realized with insulated layer (Fig. 4(d)). Figure 4(e) compares the long-term cycling performances of these three separators. Note rate performance was first tested during the cell measurements (Fig. 4(b)), and then the long-cycling testing at 0.5 C was performed. Compared with the PP and the PP-BN, the PP-BNNS separator cell delivers higher specific capacity of $750.1 \text{ mAh}\cdot\text{g}^{-1}$. The capacity only decayed to 3.6% of its initial value ($723.2 \text{ mAh}\cdot\text{g}^{-1}$) after 200 cycles, which is 8.2% and 15.7% lower in comparison with the PP-BN and the PP separators, accompanying with the average decays of 0.018%, 0.041% and 0.078% per cycle, respectively. Moreover, the cell with the PP-BNNS separator shows a high average CE of 99.7% during long cycling. The excellent cyclic and rate performances are highly competitive when compared with other carbon electrodes based on Celgard PP coated separators (see Table S5 in the ESM for details), which are mainly attributed to the physical blocking effect of BNNSs and their Lewis acid sites induced chemically adsorption, which improved the capacity and utilization of sulfur. To better understand the function of the PP-BNNS separator in Li-S batteries, the cell was disassembled to reveal its protection to Li metal anode after being cycled at 0.5 C for 200 cycles. We also tested the cycling capacities of the cells with these three separators within different temperatures at 1 C , as shown in Fig. S10 in the ESM. The cell with the PP-BNNS separator could operate normally and kept the highest capacity compared with the PP and PP-BN separator at different temperature due to the enhanced thermal properties, while the cell with PP separator failed with a few cycles under the identical conditions (Fig. S10(b) in the ESM). Additionally, the SEM images of the PP, PP-BN, PP-BNNS separators after cycled at 0.5 C were supplied in Fig. S11 in the ESM, indicating that the PP-BNNS separator is highly stable and restrain the migration of polysulfides. Meanwhile, as shown in Fig. S12 in the ESM, a lot of cracks and Li dendrites were formed on the Li metal anode surface with the PP and the PP-BN separator after cycling, while a relatively smooth surface was obtained with the PP-BNNS separator, suggesting that the BNNSs not only restrict the shuttle effect, but also can effectively protect Li metal anode from dendrites growth. Therefore, the PP-BNNS is a promising separator to high-energy safer Li metal batteries because of its high thermal ability, high ionic conductivity and high mechanical strength.

4 Conclusions

In conclusion, we demonstrated an additive free, rapid, and high-efficient SALEP method to produce high-quality BNNSs. The high feeding (2 g) and the high record of yield (72.5% after 8 cycles) make this method promising for scalable production of BNNSs. We systemically investigated the role of several factors (e.g., ultrasonic frequency/power, sonication time, solvent and cyclic number) in the yield of BNNSs and the high-record exfoliation efficiency of $10.2 \text{ }\mu\text{g}\cdot\text{h}^{-1}$ was achieved at an optimal condition (i.e., 30 kHz , 315 W and IPA/water ratio of 40/60). Importantly, the BNNSs modified separator exhibits high specific and long-term cyclic capacity in Li-S batteries. Together with low cost and high exfoliation efficiency, this method provides us a simple strategy for high-quality and

few-layered BNNs with great potential in energy storage and conversion applications.

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Electronic Supplementary Material: Supplementary material (HSP, and R_a value calculation; SEM images and their particle size distribution histograms of BNNs; the h-BN feeding and exfoliation efficiency of the reported SALEP and other methods; high-resolution XPS scan and atomic composition of the bulk h-BN and the BNNs; Young's modulus; the contact angle measurements; ionic conductivity; the voltage profiles of the Li symmetric cells; TGA curve, XRD pattern and SEM images of the KB/S cathode; the cycling capacities at wide temperature; the cyclic and rate performances of the Li-S batteries; SEM images of the separators after cycled; SEM images of the Li metal anode) is available in the online version of this article at <https://doi.org/10.1007/s12274-020-3245-3>.

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